BEU CONNECT

Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2018

Time: 3 hours

Code: 031506 *Subject: Electromagnetic Field Theory* *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Fill in the blanks any seven of the following: **[14]**

(a) VSWR of a matched load is _____

(b) Normal component of magnetic flux density is _____ across the boundary.

(c) Intrinsic impedance of lossless medium is _____

(d) If the current flowing in the two wires is in the same direction, then there will be force of _____

(e) Divergence of velocity of water well within the surface of water is _____

(f) The product of short-circuited impedance and open-circuited impedance of a transmission line is _____

(g) The input impedance of a resonant section of transmission line is given by _____

(h) Brewster angle is given by _____

(i) Surface impedance of a good conductor is _____

(j) Power loss in a plane conductor is _____

Q.2 Solve both questions : **[14]**

(a) Discuss uniqueness theorem in detail.

(b) Find the capacitance between two spheres whose separation d is very much larger than their radii R.

Q.3 Solve both questions : **[14]**

(a) Express Laplacian operator in curvilinear coordinate system and find $\nabla^2 V$, where $V=10\, r\sin^2\theta\cos\varphi$.

(b) Discuss boundary condition at the interface between two dielectric mediums and find the magnitude of D and E in one medium as compared to another medium.

Q.4 Solve all questions : **[14]**

(a) Find the energy stored in a magnetic field.

(b) Discuss Stokes' law.

(c) Discuss magnetic vector potential.

Q.5 Solve all questions : **[14]**

(a) In a medium characterized by $\sigma=0$, $\mu=\mu_0$ and $\varepsilon_r=4$, E is given by $E=20\sin(10^8 t-\beta z)y$V/m. Calculate H and β.

(b) Discuss continuity equation for current.

(c) Discuss analogies between electric and magnetic fields.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store

BEU CONNECT

Q.6 Solve both questions : **[14]**

(a) Find the ratio of E and H in a uniform plane wave in a lossless medium.

(b) Using $\nabla \cdot D = \rho$, Ohm's law and equation of continuity, show that if at any instant a charge density ρ existed within a conductor, it would decrease to $1/e$ times this value in a time ε/σ sec. Calculate this time for copper conductor.

Q.7 Solve this question : **[14]**

The electric field strength of a uniform plane electromagnetic wave in free space is 1V/m and the frequency is 300 MHz. If a very large thick flat copper plate is placed normal to the direction of wave propagation, determine (a) the electric field strength and magnetic field strength at the surface of the plate, (b) the depth of penetration, (c) the conduction current density at the surface, (d) conduction current density at a distance of 0.01 mm below the surface, (e) the linear current density, I_s, (f) the surface impedance, (g) the power loss per square meter of surface area. For copper $\sigma = 5.8 \times 10^7$, $\varepsilon = \varepsilon_0$ and $\mu = \mu_0$.

Q.8 Solve both questions : **[14]**

(a) Explain Poynting theorem.

(b) A short vertical transmitting antenna erected on the surface of a perfectly conducting earth produces effective field strength $E_{eff} = E_{\theta eff} = 100 \sin\theta mV/m$ at points a distance of one mile from the antenna. Compute the Poynting vector and total power radiated.

Q.9 Solve both questions : **[14]**

(a) Find the quality factor of a resonant transmission line section.

(b) Discuss quarter wave line as transformer.

BEU CONNECT
LEARN • GROW • SUCCEED
Your Engineering Journey, Simplified
PYQs • SYLLABUS • RESULTS
AI DOUBT SOLVER • ATTENDANCE • GATE MOCK TESTS
Free • Fast • Made by students
GET IT ON Google Play
Scan QR to download
Search BEU Connect on Play Store